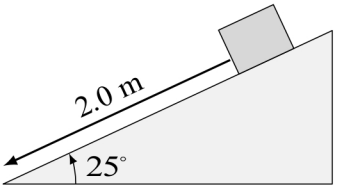
PHYS-1310
EXAM 02
Given Time: 2 hrs

Name: _____ Student ID: _____

Problems 1 and 2: A 3.0 kg block slides 2.0 m down a frictionless ramp inclined at 25°.



1. The work done by the normal force on the block over the 2.0 m distance is

a. 25 J

b. 29 J

c. 53 J

d. 59 J

e. 0

2. The work done by gravity on the block over the 2.0 m distance is

a. 25 J

b. 29 J

c. 53 J

d. 59 J

e. 0

3. A 900 kg elevator is moving up at a constant speed. In 20 s the elevator rises 14 m. The average power of the elevator's motor over this interval is

a. 630 W

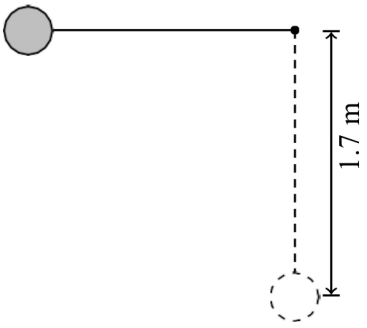
b. 1700 W

c. 220 W

d. 6200 W

e. 3100 W

4. A 0.90 kg ball on the end of a 1.7 m long string is released from the horizontal. It descends along its circular path. As it passes though the bottom, the tension in the string is



a. 8.8 N

b. 18 N

c. 26 N

d. 43 N

e. 54 N

5. A spring launches a 4.0 kg block across a frictionless floor. The spring constant is k = 3600 N/m and the spring is initially compressed 0.20 m. The launch speed of the block is

a. 6.0 m/s

b. 9.5 m/s

c. 13 m/s

d. 8.5 m/s

e. 19 m/s

6. From rest, a 3.0 kg block slides down a hill of height 2.5 m and then across a floor. If the block-floor friction coefficient is $\mu_k = 0.43$, how far does the block slide across the floor?

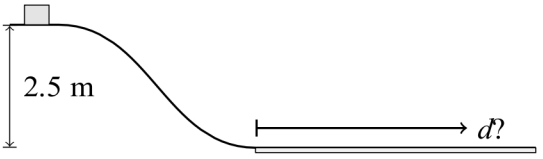
a. 2.5 m

b. 7.3 m

c. 8.2 m

d. 4.3 m

e. 5.8 m



7. These four uniform tiles are identical. Each has a side length of 1.0 m. What is y_{com} for these tiles?

a. 1.2 m

b. 0.25 m

c. 0.50 m

d. 0.75 m

e. 1.0 m

8. On a plane of ice, a heavy puck collides with a light puck. Which choice describes the center-of-mass (COM) trajectory of this 2-puck system?

a. How the collision alters the COM trajectory depends on the relative masses of the pucks.

b. How the COM trajectory is altered depends on the extent to which the collision is inelastic.

c. The COM moves with constant velocity showing no sign of the collision.

d. The degree the collision is glancing versus head-on affects how the COM trajectory is altered.

e. The COM trajectory can be altered by multiple factors including at least two of the above.

9. The depicted time-dependent force acts on a 2.7 kg block. At $t = 0$, $v_{\text{block}\,x} = -1.2$ m/s. At $t = 3.0$ s, $v_{\text{block}\,x} = 3.6$ m/s. What is F_{max}?

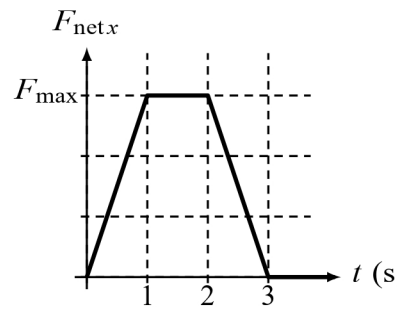
A. 3.2 N

B. 2.2 N

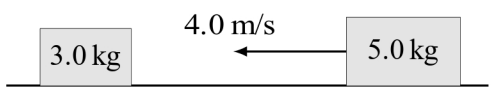
C. 6.5 N

D. 5.2 N

E. 4.3 N



Problems 10 and 11: A 5.0 kg block with an incoming speed of 4.0 m/s strikes a target 3.0 kg block. The floor is frictionless.



10. If the collision is completely inelastic, the speed of the stuck-together blocks after the collision is

a. 4.0 m/s

b. 8.0 m/s

c. 7.3 m/s

d. 2.5 m/s

e. 5.0 m/s

11. If the collision is elastic, the speed of the 3.0 kg target block after the collision is

a. 4.0 m/s

b. 8.0 m/s

c. 7.3 m/s

d. 2.5 m/s

e. 5.0 m/s

12. These two pieces of clay have different masses but the same speed of 10 m/s. They collide forming a single 5.0 kg clay blob. The speed of the 5.0 kg clay blob is

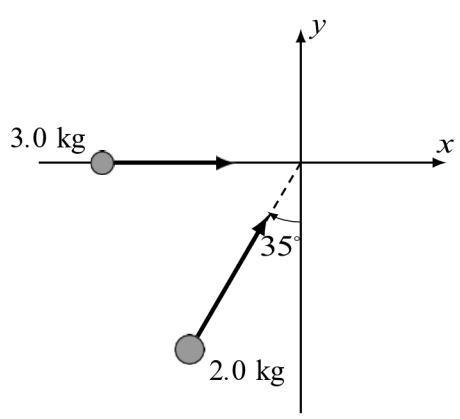
A. 8.9 m/s

B. 7.5 m/s

C. 9.6 m/s

D. 10 m/s

E. 6.8 m/s



13. A wheel rotating at 5.0 rev/s has a kinetic energy of 1800 J. The wheel's rotational inertia I is

a. 140 kg m^2

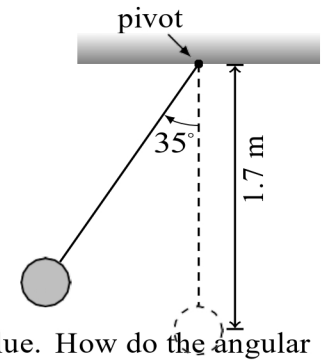
b. 5700 kg m^2

c. 3.6 kg m^2

d. 57 kg m^2

e. 110 kg m^2

14. A 0.4 kg ball swings from the end of a string of length 1.7 m. When the string makes a 35° angle with the vertical, the magnitude of the torque on the ball about the pivot is



a. 2.3 N·m

b. 0

c. 3.8 N·m

d. 3.2 N·m

e. 5.5 N·m

15. A spinning star collapses and its rotational inertia shrinks to 1/3 of its initial value. How do the angular speeds before and after the collapse compare?

a. Not enough information is given.

b. $\omega_f = \omega_0$

c. $\omega_f = 3\omega_0$

d. $\omega_f = \sqrt{3\omega_0}$

e. $\omega_f = 9\omega_0$

16. A 0.20 kg mass oscillates on the end of a spring according to x = (0.060 m) cos[(3.4 rad/s) t]. The mechanical energy of this oscillator is

a. 0.020 J

b. 0.24 J

c. 0.12 J

d. 0.0042 J

e. 0.0061 J

17. A block that hangs from a spring is pulled down slightly and released. Over the span of the block's oscillations, where do v_{max} and a_{max} occur?

a. v_{max} at the middle, a_{max} at the top and bottom.

b. v_{max} at the top and bottom, a_{max} at the middle.

c. v_{max} and a_{max} both in the middle.

d. v_{max} and a_{max} both at the top and bottom.

e. v_{max} at the middle, a_{max} just at the bottom.

18. A mass on the end of a string swings with a small amplitude. If the string's length of the string is doubled, $f_{\text{long string}}/f_{\text{short string}}$ is

a. 0.707

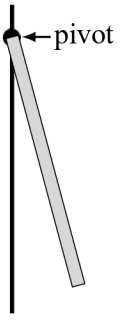
b. 1.00

c. 2.00

d. 0.500

e. 1.41

19. A uniform rod of length 1.00 m is mounted to a pivot at one end ($I = \frac{1}{3}ML^2$), the period of oscillation for small amplitudes is



a. 1.42 s

b. 1.16 s

c. 0.82 s

d. 2.01 s

e. 1.64 s

Problems 20 and 21: A transverse wave is described by

$$y(x, t) = (0.10 \text{ m}) \cos[(5.0 \text{ rad/m})x - (13 \text{ rad/s})t].$$

20. The speed of this traveling wave is

a. 4.2 m/s

b. 2.6 m/s

c. 4.5 m/s

d. 1.3 m/s

e. 3.9 m/s

21. For a single point on the string, this point's maximum transverse speed is

a. 4.2 m/s

b. 2.6 m/s

c. 4.5 m/s

d. 1.3 m/s

e. 3.9 m/s

22. A worker pushes a wheelbarrow with a horizontal force of 50 N on level ground over 5.0 m. If a friction force of 43 N acts on the wheelbarrow in a direction opposite that of the worker, what work is done on the wheelbarrow by the worker?

A. 250 J
B. 215 J
C. 35 J
D. 10 J
E. None of those answers is correct.

23. A cart is set rolling across a level table, at the same speed on every trial. If it runs into a patch of sand, the cart exerts on the sand an average horizontal force of 6 N and travels 6 cm through the sand as it comes to a stop. If instead the cart runs into a patch of gravel on which the cart exerts an average horizontal force of 9 N, how far into the gravel will the cart roll before stopping?

A. 9 cm
B. 6 cm
C. 4 cm
D. 3 cm
E. none of those answers

24. Bullet 2 has twice the mass of bullet 1. Both are fired so that they have the same speed. If the kinetic energy of bullet 1 is K, is the kinetic energy of bullet 2 (a) $0.25K$,

A. $0.5K$,
B. $0.71K$,
C. K, or
D. $2K$?

25. A certain spring that obeys Hooke's law is stretched by an external agent. The work done in stretching the spring by 10 cm is 4 J. How much additional work is required to stretch the spring an additional 10 cm?

A. 2 J
B. 4 J
C. 8 J
D. 12 J
E. 16 J

26. A cart is set rolling across a level table, at the same speed on every trial. If it runs into a patch of sand, the cart exerts on the sand an average horizontal force of 6 N and travels 6 cm through the sand as it comes to a stop. If instead the cart runs into a patch of flour, it rolls an average of 18 cm before stopping. What is the average magnitude of the horizontal force the cart exerts on the floor?

A. 2 N
B. 3 N
C. (6 N
D. 18 N
E. none of those answers

27. What average power is generated by a 70.0-kg mountain climber who climbs a summit of height 325 m in 95.0 min?

A. 39.1 W
B. 54.6 W
C. 25.5 W
D. 67.0 W
E. 88.4 W

28. A 2-kg object moving to the right with a speed of 4 m/s makes a head-on, elastic collision with a 1-kg object that is initially at rest. The velocity of the 1-kg object after the collision is

A. greater than 4 m/s,
B. less than 4 m/s,
C. equal to 4 m/s,
D. zero, or
E. impossible to say based on the information provided.

29. A 5-kg cart moving to the right with a speed of 6 m/s collides with a concrete wall and rebounds with a speed of 2 m/s. What is the change in momentum of the cart?

A. 0
B. 40 kg - m/s
C. -40 kg -m/s
D. -30 kg -m/s
E. -10 kg - m/s

30. A 57.0-g tennis ball is traveling straight at a player at 21.0 m/s. The player volleys the ball straight back at 25.0 m/s. If the ball remains in contact with the racket for 0.060 0 s, what average force acts on the ball?

A. 22.6 N
B. 32.5 N
C. 43.7 N
D. 72.1 N
E. 102 N

31. A 10.0-g bullet is fired into a 200-g block of wood at rest on a horizontal surface. After impact, the block slides 8.00 m before coming to rest. If the coefficient of friction between the block and the surface is 0.400, what is the speed of the bullet before impact?

A. 106 m/s
B. 166 m/s
C. 226 m/s
D. 286 m/s
E. none of those answers is correct

32. Two particles of different mass start from rest. The same net force acts on both as they move over equal distances. How do their final kinetic energies compare?

A. The particle of larger mass has more kinetic energy.
B. The particle of smaller mass has more kinetic energy.
C. The particles have equal kinetic energies.
D. Either particle might have more kinetic energy.

33. Consider an object on a rotating disk a distance *r* from its center, held in place on the disk by static friction. Which of the following statements is *not* true concerning this object?

A. If the angular speed is constant, the object must have constant tangential speed.
B. If the angular speed is constant, the object is not accelerated.
C. (The object has a tangential acceleration only if the disk has an angular acceleration.
D. If the disk has an angular acceleration, the object has both a centripetal acceleration and a tangential acceleration.
E. The object always has a centripetal acceleration except when the angular speed is zero.

34. An ice skater starts a spin with her arms stretched out to the sides. She balances on the tip of one skate to turn without friction. She then pulls her arms in so that her moment of inertia decreases by a factor of 2. In the process of her doing so, what happens to her kinetic energy?

A. It increases by a factor of 4.
B. It increases by a factor of 2.
C. It remains constant.
D. It decreases by a factor of 2.
E. It decreases by a factor of 4.

35. Which of the following statements is *not* true regarding a mass–spring system that moves with simple harmonic motion in the absence of friction?

A. The total energy of the system remains constant.
B. The energy of the system is continually transformed between kinetic and potential energy.
C. The total energy of the system is proportional to the square of the amplitude.
D. The potential energy stored in the system is greatest when the mass passes through the equilibrium position.
E. The velocity of the oscillating mass has its maxi- mum value when the mass passes through the equilibrium position

36. An object–spring system moving with simple harmonic motion has an amplitude *A*. When the kinetic energy of the object equals twice the potential energy stored in the spring, what is the position *x* of the object?
A. *A*
B. 1/3A
C, $A/\sqrt{3}$
D. 0
E. none

37. If one end of a heavy rope is attached to one end of a lightweight rope, a wave can move from the heavy rope into the lighter one. What happens to the speed of the wave?

A. It increases.
B. It decreases.
C. It is constant.
D. It changes unpredictably.

38. The distance between two successive peaks of a sinusoidal wave traveling along a string is 2 m. If the frequency of this wave is 4 Hz, what is the speed of the wave?

A. 4 m/s
B. 1 m/s
C. 8 m/s
D. 2 m/s
E. impossible to answer from the information given

39. Two traveling waves: and

$y_1 = A \sin[k(x - vt)]$

$y_2 = A \sin[k(x + vt)]$

are superposed on the same string. The distance between the adjacent nodes of the resulting standing wave is:

A. π/k
B. vt/π
C. $vt/2\pi$
D. $\pi/2k$
E. $2\pi/k$

40. A weight suspended from an ideal spring oscillates up and down. If the amplitude of the oscillation is doubled, the period will:
A. remain the same.
B. increase by a factor of $\sqrt{2}$.
C. double.
D. halve.
E. decrease by a factor of $\sqrt{2}$.

41. A particle moves in simple harmonic motion according to $x = 2 \cos(50t)$, where x is in meters and t is in seconds. Its maximum velocity in m/s is
A. 100
B. 100 sin(50t)
C. 100 cos(50t)
D. 200
E. none of these

42. The rotational inertia of a uniform thin rod about its end is $ML^2/3$, where M is the mass and L is the length. Such a rod is hung vertically from one end and set into small amplitude oscillation. If L = 1.0 m and M = 200 g, this rod will have the same period as a simple pendulum of length:

A. 67 cm
B. 33 cm
C. 50 cm
D. 100 cm
E. 150 cm

43. Sinusoidal water waves are generated in a large ripple tank. The waves travel at 20 cm/s and their adjacent crests are 5.0 cm apart. The time required for each new whole cycle to be generated is:

A. 0.25 s
B. 100 s
C. 4.0 s
D. 2.0 s
E. 0.5 s

44. What is the x-coordinate of the center of mass of an object consisting of the following point masses; M at (0,0), 2M at (10,0), 3M at (10,10) and 4M at (20,0)?

A. 13
B. 14
C. 15
D. 10
E. 12

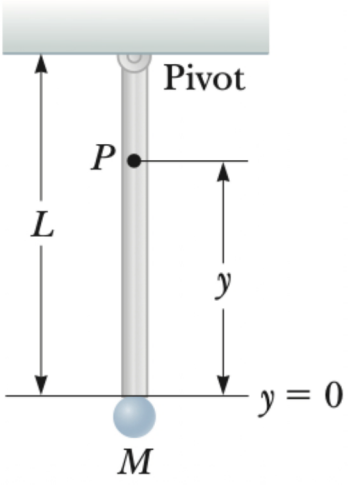
45. A Simple Harmonic Oscillator has a displacement of x = 3 cos(2t). What is the maximum acceleration?

A. 12
B. 6
C. 3
D. 9
E. 18

46. A spring with a 3-kg mass is held stretched 0.6 m beyond its natural length by a force of 20 N. If the spring begins at its equilibrium position but a push gives it an initial velocity of 1.2 m/s, find the position of the mass after *t* seconds.

A. x(t) = 0.36 sin (10t/3)
B. x(t) = 0.6 sin (8t/3)
C. x(t) = 1.2 cos (10t/3)
D. x(t) = 0.36 cos (10t/3)

47. A small ball of mass *M* is attached to the end of a uniform rod of equal mass *M* and length *L* that is pivoted at the top Determine this the period for *L* =2.00 m.



A. 2.68 s
B. 3.28 s
C. 6.3 s
D. 26.8 s

48. A transverse wave on a string is described by the wave **function**

$$y = 0.120 \sin\left(\frac{\pi}{8}x + 4\pi t\right)$$

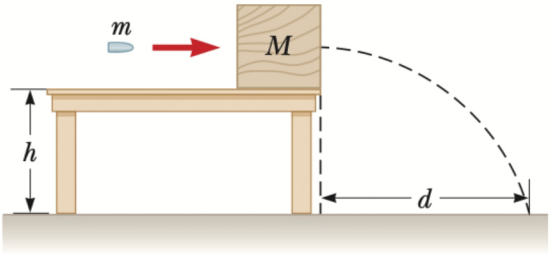
where x and y are in meters and t is in seconds. Determine the wavelength, the period, and the speed of propagation of this wave are

A. 16m, 0.5s, 32m/s
B. 1.6m, 1s, 16m/s
C. 32m,0.5s, 16m
D. 160m, 2s, 64m/s

49. A sinusoidal wave in a string is described by the wave function $y = 0.150 \sin(0.800x - 50.0t)$ where x and y are in meters and t is in seconds. The mass per length of the string is 12.0 g/m. The maximum transverse acceleration of an element of this string.

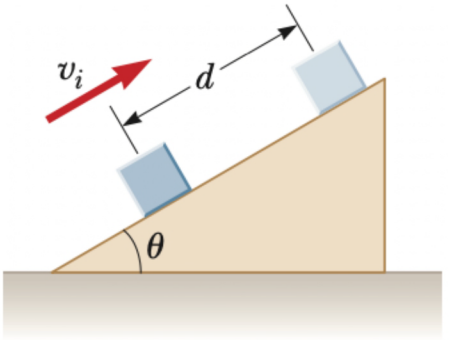
A. 375m/s^2
B. 300 m/s^2
C. 175m/s^2
D. 300m/s^2

50. A bullet of mass m 5 8.00 g is fired into a block of mass M = 250 g that is initially at rest at the edge of a table of height h =1.00 m. The bullet remains in the block, and after the impact the block lands d = 2.00 m from the bottom of the table. Determine the initial speed of the bullet.



A. 100m/s
B. 143m/s
C. 123m/s
D. 100m/s

51. A 5.00-kg block is set into motion up an inclined plane with an initial speed of v = 8.00 m/s . The block comes to rest after traveling d =3.00 m along the plane, which is inclined at an angle of q = 30.0° to the horizontal. For this motion What is the coefficient of kinetic friction?



A. 0.5
B. 0.75
C. 0.679
D. 6.79

PHYS-1310
Exam 02 Answer Sheet

Name: _____ Student ID: _____

Multiple Choice

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